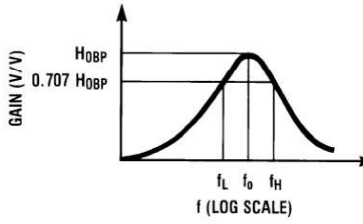


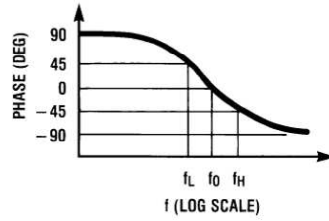
Annexe

Filtres réalisables avec le LMF100

$$H_{BP}(s) = \frac{H_{OBP} \frac{\omega_0}{Q} s}{s^2 + \frac{s\omega_0}{Q} + \omega_0^2}$$



(a)



(b)

$$Q = \frac{f_0}{f_H - f_L}; f_0 = \sqrt{f_L f_H}$$

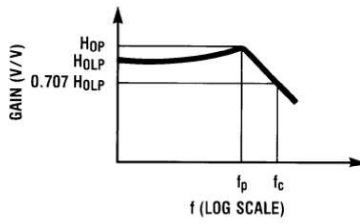
$$f_L = f_0 \left(\frac{-1}{2Q} + \sqrt{\left(\frac{1}{2Q}\right)^2 + 1} \right)$$

$$f_H = f_0 \left(\frac{1}{2Q} + \sqrt{\left(\frac{1}{2Q}\right)^2 + 1} \right)$$

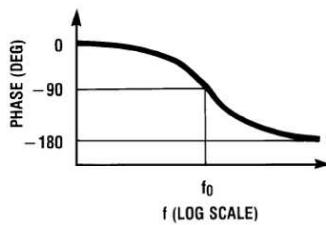
$$\omega_0 = 2\pi f_0$$

FIGURE 1. 2nd-Order Bandpass Response

$$H_{LP}(s) = \frac{H_{OLP} \omega_0^2}{s^2 + \frac{s\omega_0}{Q} + \omega_0^2}$$



(a)



(b)

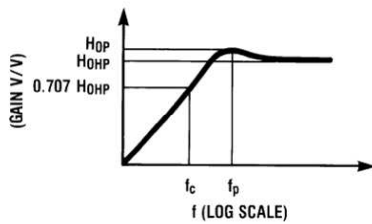
$$f_c = f_0 \times \sqrt{\left(1 - \frac{1}{2Q^2}\right) + \sqrt{\left(1 - \frac{1}{2Q^2}\right)^2 + 1}}$$

$$f_p = f_0 \sqrt{1 - \frac{1}{2Q^2}}$$

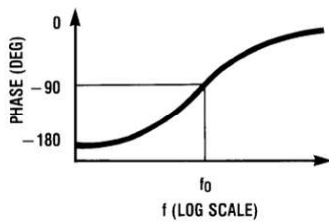
$$H_{OP} = H_{OLP} \times \frac{1}{\frac{1}{Q} \sqrt{1 - \frac{1}{4Q^2}}}$$

FIGURE 2. 2nd-Order Low-Pass Response

$$H_{HP}(s) = \frac{H_{OHP} s^2}{s^2 + \frac{s\omega_0}{Q} + \omega_0^2}$$



(b)



$$f_c = f_0 \times \left[\sqrt{\left(1 - \frac{1}{2Q^2}\right) + \sqrt{\left(1 - \frac{1}{2Q^2}\right)^2 + 1}} \right]^{-1}$$

$$f_p = f_0 \times \left[\sqrt{1 - \frac{1}{2Q^2}} \right]^{-1}$$

$$H_{OP} = H_{OHP} \times \frac{1}{\frac{1}{Q} \sqrt{1 - \frac{1}{4Q^2}}}$$

FIGURE 3. 2nd-Order High-Pass Response